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Computer Vision – Short Answer Questions

1. What is Eigenfaces?

Eigenfaces is a face recognition technique based on **Principal Component Analysis (PCA)**. It represents a face as a combination of important facial features called **eigenvectors or eigenfaces**. A new face is recognized by comparing its features with stored face features.

2. Define PCA in Image Processing.

Principal Component Analysis (PCA) is a dimensionality reduction technique used to reduce the number of features in an image while retaining the most important information. It transforms image data into a smaller set of principal components.

3. What is Contour-Based Shape Representation?

Contour-based shape representation describes an object's shape using its **boundary or outline**. Features such as boundary points, curvature, chain codes, and Fourier descriptors can be used to represent the shape.

4. What is Region-Based Shape Representation?

Region-based shape representation describes a shape using **all pixels inside the object's boundary**. It considers properties such as area, moments, texture, and internal structure.

5. How Does an Active Contour (Snake) Work?

An **active contour or snake** is a flexible curve that moves toward an object's boundary. It is influenced by:

- Internal energy, which keeps the contour smooth
- External energy, which attracts the contour toward image edges

The contour gradually adjusts until it fits the object boundary.

6. How Does a Level Set Method Work?

The **level set method** represents a contour as a mathematical function. The contour evolves over time according to image information. It can naturally handle:

- Object splitting
- Object merging
- Complex shape changes

It is commonly used for image segmentation.

7. What is the Hough Transform?

The **Hough Transform** is a feature extraction technique used to detect geometric shapes such as:

- Lines
- Circles
- Ellipses

It transforms image points into a parameter space and finds peaks corresponding to possible shapes.

8. How Are Fourier Descriptors Used in Shape Analysis?

Fourier descriptors represent a shape boundary using the **Fourier Transform**. The boundary coordinates are converted into frequency components. Low-frequency components represent the overall shape, while high-frequency components represent fine details.

They are useful for shape matching and recognition.

9. How Are Wavelet Descriptors Used in Shape Analysis?

Wavelet descriptors analyze a shape at **multiple resolutions or scales**. They capture both global and local shape information. They are useful for detecting fine shape details and analyzing shapes with different sizes.

10. What is a Particle Filter?

A **particle filter** is a tracking algorithm that represents the possible state of an object using many particles. Each particle represents a possible position and movement of the object.

The particles are updated according to observations, making particle filters useful for tracking objects under:

- Occlusion
 - Noise
 - Uncertain movement
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11. How is Foreground–Background Separation Done in Surveillance Systems?

Foreground–background separation is performed by comparing the current video frame with a background model.

Basic Steps:

1. Build a background model.
2. Compare the current frame with the background.
3. Identify significantly changed pixels as foreground.
4. Remove noise using filtering and morphological operations.
5. Update the background model over time.

Methods such as **Gaussian Mixture Models (GMM)** are commonly used.

12. How Do Multi-Camera Systems Track Pedestrians?

Multi-camera systems track pedestrians by:

1. Detecting pedestrians in each camera.
2. Extracting appearance features.
3. Tracking movement within each camera.
4. Comparing person features across cameras.
5. Assigning the same identity to matching pedestrians.

Person Re-Identification (Re-ID) and camera calibration are commonly used for cross-camera tracking.

13. What is a Convolutional Neural Network (CNN)?

A **Convolutional Neural Network (CNN)** is a deep learning model mainly used for image processing and computer vision.

It contains:

- Convolution layers for feature extraction
- Activation functions
- Pooling layers for dimensionality reduction
- Fully connected layers for classification

CNNs are widely used for image classification, object detection, and face recognition.

14. How Does a Vision Transformer (ViT) Work?

A **Vision Transformer (ViT)** divides an image into small patches. Each patch is converted into a numerical embedding and processed using transformer layers with a **self-attention mechanism**.

Self-attention helps the model understand relationships between different parts of an image.

Image → Patches → Embeddings → Transformer Encoder → Classification Output

15. What is a Generative Adversarial Network (GAN)?

A **Generative Adversarial Network (GAN)** consists of two neural networks:

- **Generator:** Creates synthetic data or images.
- **Discriminator:** Determines whether an image is real or generated.

The two networks compete with each other, causing the generator to gradually produce more realistic images.

16. How is a Support Vector Machine (SVM) Used for Image Classification?

In image classification, features are first extracted from images using methods such as:

- HOG
- SIFT
- CNN features

These features are given to an **SVM classifier**. The SVM finds the best decision boundary that separates different image classes.

17. How Are Hidden Markov Models (HMM) Applied in Activity Recognition?

An **HMM** models an activity as a sequence of hidden states.

For example, the activity **walking** may consist of different movement states. Observations from video frames or sensors are used to estimate the most likely activity sequence.

HMMs are useful for recognizing:

- Walking
 - Running
 - Sitting
 - Gestures
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18. What is a Gaussian Mixture Model (GMM)?

A **Gaussian Mixture Model (GMM)** is a probabilistic model that represents data as a combination of multiple Gaussian distributions.

Each Gaussian component represents a different group or pattern in the data.

In computer vision, GMM is commonly used for:

- Background modeling
 - Foreground detection
 - Image segmentation
 - Data clustering
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19. How is Graph-Based Learning Applied to 3D Shape Correspondence?

In graph-based learning, a 3D shape is represented as a graph where:

- **Vertices** represent points on the shape.
- **Edges** represent relationships between neighboring points.

Graph-based algorithms compare structural and geometric relationships between two 3D shapes to find corresponding points.

This is useful for:

- 3D object matching
- Shape correspondence
- Medical shape analysis

- 3D model recognition
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20. How is Road and Lane Detection Performed in Autonomous Driving?

Road and lane detection is performed using cameras and computer vision algorithms.

Basic Steps:

1. Capture the road image using a front-facing camera.
2. Convert the image to grayscale or another suitable color space.
3. Apply image enhancement and noise reduction.
4. Select the road **Region of Interest (ROI)**.
5. Detect edges using the **Canny Edge Detector**.
6. Detect lane lines using the **Hough Transform**.
7. For curved lanes, use polynomial curve fitting.
8. Use previous frames to improve temporal stability.
9. Overlay the detected lanes on the original image.

Modern autonomous vehicles may also use:

- CNN-based lane detection
- Semantic segmentation
- Vision Transformers
- LiDAR
- Radar
- Sensor fusion

Pipeline:

Camera Input

→

Preprocessing

→

ROI Selection

→

Edge Detection

→

Lane Detection

→

Curve Fitting

→

Lane Tracking

→

Autonomous Driving Decision